PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

E-Commerce Analytics Dashboard

"Analyzing Customer’s Sales Behaviour
and Tracking Business Process"

A Capstone Project

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Approval Sheet

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Acknowledgement

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TABLE OF CONTENTS

Title page ----------I

Approval Sheet ----------II

Acknowledgement ----------III

Table of Contents ----------IV,V,VI

Abstract ----------VII,VIII

CHAPTER I

Introduction ----------9

Background of the Study ----------10,11

Statement of the Problem ----------11,12

Objectives of the Study ----------12,13

Significance of the Study ----------13,14

Scope and limitations ----------14,15

Definition of Terms ----------16,17

CHAPTER II

Overview ----------18,19

Traditional vs Modern System----------19

Literature Review ----------20,21,22,23

Case Studies ----------20,21,22,23

CHAPTER III

Methodology ----------25

Research Design ----------26,27

Model Use ----------27,28,29,30

Use-case Diagram ----------33

Data Flow Diagram ----------34

Entity Relationship Diagram ----------34

Survey Questions ----------35,36,37,38

CHAPTER IV

Result and Discussion ----------39

Systems and Features ----------39,40

Implementation Challenges ----------40,41

User Feedback ----------41

Comparison to other Systems ----------42

CHAPTER V

Conclusion and Recommendations ----------43

Conclusion ----------43

Recommendations ----------44,45

Final Thoughts ----------46

Appendices ----------47

Screenshots ----------47,48

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

Curriculum Vitae ----------49,50,51,52

Grammarian Certificate ----------

Abstract

The study entitled **E-Dash Analyzing Customer's Sales Behaviour and Tracking Business Process** focuses on the development and evaluation of analyzing customers behaviour when it comes to sales and tracking their business process in sales. It is an e-commerce dashboard mobile application that integrates modern technologies such as Data Analytics Tools, Mobile Application, Android Studio (IDE). The main goal of the project is to create and implement an advanced E-commerce Dashboard Application that implements the tracking, overseeing, and analyzing customer information in the use of **E-Dash Application.**

The application was designed to provide a convenient and interactive platform for business owners to monitor their sales and to analyze their customer's behaviour when it comes to sales and monitor the product movement and to know what are the top products within the day, months, and a year based on their sales. It integrates features such as Top Product Dashboard, Showing Product History, and Weekly, Monthly sales Inventory and Monitoring stocks inventor**y** to enhance engagement and build owner's trust. The study employed both qualitative and quantitative methods, including interviews, observation and user testing, to assess the system's usability, functionality, and business operation effectiveness.

Results showed that Ecommerce Dashboard plays a significant role in shaping customer perceptions and increasing sales intent. Respondents indicated that clear Product movement, Showing Top Product accordingly, and owner's satisfaction were the most

factors that motivated them to gain their trust. The system’s user interface and inventory process were rated highly satisfactory, confirming that a well structured ecommerce platform combined with effective sales strategies can positively affect costumers decisions.

In conclusion, the developed Ecommerce Dashboard application successfully demonstrates the relationship between business owners and customers' sales behaviour. The integration of marketing tools within the system not only improves customer engagement and satisfaction but also provides a practical model for small businesses seeking to establish an online presence and expand their inventory sales reach.

Chapter 1

Introduction of the Study

In today's digital economy, e-commerce has become a dominant impact on how businesses operate and interact with customers. Companies rely heavily on real-time data analytics to monitor sales, track business processes, and understand customer data. However, many businesses are still struggling with inefficient tracking systems, and limited access to actionable insights. Without an efficient approach, decision-making processes become reactive rather than proactive, leading to inefficiencies in inventory management, sales monitoring, understanding customers' insights and customer relationships.

To address these challenges, businesses require an intelligent, data driven solution that collects and automates key operational metrics. An E-commerce Analytics Dashboard can bridge this gap by offering real-time tracking, analytics, and visualization tools to enhance efficiency and optimize decision-making.

The main objective of this capstone project is to create and implement an advanced E-commerce Dashboard Application that implements the tracking, overseeing, and analyzing customer information. This solution aims to automate vital processes including sales monitoring, process monitoring, and identify customer needs, allowing businesses to make informed decisions to enhance their business operation efficiently.

Background of the Study

Large data companies need to process and comprehend. The e-commerce systems of today can access huge amounts of data on consumers, what they've viewed, what they've purchased, and this can be invaluable when figuring out what customers need. E-commerce platforms produce a huge amount of customer data, including customer shopping behaviour, purchase history, and customer needs, and this data can be quite insightful into customer needs and business performance. But small and medium-size businesses are not able to take advantage of this data due to inefficient tracking systems and the inability to process unprocessed data into useful insights. The e-commerce industry has grown very rapidly in the past two years, revolutionizing the way businesses perform their commerce and consumers engage with goods and services. Even though the online environment has allowed companies to interact with consumers differently, there also exists different difficulties when it comes to the

In summary, e-commerce businesses have relied on traditional methods, such as spreadsheets and reports, for tracking their sales, customer behaviour, and campaign performance. The need for a larger and more automatic process has led to the creation of E-commerce Analytics Dashboards, with real-time tracking, data visualization, and more enhanced analytics tools for data-driven decision-making for companies.

The objective of this research is to overcome these deficiency with the design of an E-commerce Analytics Dashboard, automating data collection, real-time analysis, and provision of access to a simple and developed interface for business owners. The system will enable automating major processes for SMEs in the e-commerce industry, providing them with the means of data-driven decision-making for possible customer relationships and enhanced performance of the business processes.

Statement of the Problem

Many small to medium enterprises (SMEs) rely on manual tracking systems that make it difficult to analyze customer behaviour and sales data effectively. Without real-time insights, businesses struggle with inefficient decision-making, leading to challenges in sales monitoring, inventory management, and customer management.

This capstone project will aim to address the following challenges by developing an E-commerce Analytics Dashboard within the E-Dash mobile application, which will focus on real-time data visualization and analytics.

Specifically, this study seeks to answer the following

1. How can e-commerce analytics dashboards improve the accuracy in tracking the customers behaviour and sales data?

2. What impact will real-time data analytics have on the decision-making process for small-medium businesses?
3. How can an automated e-commerce analytics dashboard improve operational efficiency ?

This study seeks to address these problems by developing an advanced Ecommerce Analytics Dashboard that integrates modern technologies like Android studio, an advanced IDE that suits for developing Mobile Applications for Android platform, Java Application, and data analytics. The goal of this project is to create a mobile application that provides accurate, real-time inventory tracking, optimizes stock levels, and enhances overall efficiency, helping small to medium enterprises reduce errors, save costs, and improve customer satisfaction.

Objectives of the Study

The primary objective of this study is to design and develop an Ecommerce Analytics Dashboard that improves the accuracy, efficiency, and effectiveness of inventory management in businesses. The specific objectives of this study are as follows

- To automate analysis of customer transactions through the use of E-Dash mobile applications.
- To Develop a mobile application with an interactive dashboard to help SMEs analyze patterns and trends, improve decision-making, and enhance customer experience.

- To provide a user-friendly mobile application that is easy to navigate and accessible to business owners.

Significance of the Study

This study is significant in the reason that it can benefit small-medium businesses in the e-commerce industry, including customers, store owners who have direct access to the application. The proposed system will provide tools for data visualization, enabling business owners to develop effective marketing and financial strategies tailored to meet customer needs and goals. This in turn, is expected to enhance the profitability and business performance in the e-commerce industry.

Additionally, the system will help to reduce lost marketing sales, minimize operational costs, prevent supply deficiencies and improve the efficiency of transactions. Customers will also benefit as businesses will optimize the shopping experience, promoting customer loyalty and maintaining a strong and reputable relationship between buyers and sellers within the business industry.

The key benefits of this study include

1. **Better decision-making** through real-time data visualization and analytics can make business owners and customers informed decisions regarding inventory, stock rotation, and demand forecasting.

2. **Data visualization and analytics** enable small-medium businesses to design strategies that will be tailored to customer's sales behaviours and sales trends.
3. **Enhanced Customer Satisfaction** By ensuring that products are always available when needed, the system will help improve customer satisfaction and loyalty.

Scope and Limitation of the Study

This study focuses on the development of an Ecommerce Analytics Dashboard designed to improve the efficiency and accuracy of tracking the business process and analyze the customer's behaviour when it comes to sales. The key features of the system include inventory tracking using real-time analytics tools, and data analytics for data forecasting. The study will cover the design, development, and implementation of the system, including

1. **Data Visualization**
   - Provides visual summaries through charts, graphs, and tables for analysis.
2. **Marketing Metrics**
   - Tracks the performance of marketing information such as top sales products.
3. **Inventory Management**
   - Monitors stock levels to prevent overstocking or understocking.

The application will be designed for use in business industries such as retail, manufacturing, and small foods businesses. The scope includes the development, testing, and evaluation of the system in a real-world scenario.

Limitations

Integration Challenges

- The dashboard might face some issues integrating with some existing e-commerce platforms. E-commerce businesses often use a variety of platforms for payments, inventory management, and customer support. Integrating with these systems transitions can be difficult.

Security Concerns

OTP and Two-Factor Authentication

- Although OTP and two-factor authentication improve security, they are not reliable. User accounts are at risk, since otp received by SMS can be collected by malware, phishing scams, or SIM switching.

Data Protection

- The handling and storage of sensitive customer and business data may expose the system to potential data protection risks, including unauthorized access and data breaches

Definition of Terms

1. *E-commerce Dashboard* – An e-commerce dashboard is a tool used by e-commerce businesses to track and analyze key metrics and data related to their online sales and operations.
2. *E-commerce Analytics Dashboard* – An automated system that is designed to help online businesses to track and analyze the key data points related to customer behavior and business operations.
3. *Data Visualization Tools* – a software that is used to visualize data.
4. *Microsoft Excel* – used to format, organize and calculate data in a spreadsheet.
5. *Google Analytics Tools* – web analytics service used to provide numerous analytical tools useful for insights on website performance and marketing campaigns.
6. *Data Protection* - the process of protecting sensitive information from damage, loss or corruption.
7. *Real-time Data Analytics* - is a set of techniques for processing data as soon as it becomes available.
8. *Actionable Insights* **-** are conclusions drawn from data that can be turned directly into an action or a response.
9. *Marketing Metrics* – measure the success of marketing campaigns and show how well campaigns are tracking toward key performance indicators (KPIs).
10. *Security Concern**s*** – refers to the capacity of networks or information systems to resist unlawful or malicious actions.

11. *Small-Medium Business* **-** *maintains revenue, assets, or employee numbers below certain thresholds, usually characterized by having fewer employees and less revenue* than large corporations.
12. *Operational Efficiency* **-** a metric that measures the efficiency of profit earned as a function of operational costs.

13. *Financial Reporting* - is an accounting process that communicates financial data to external and internal stakeholders, such as shareholders.

Chapter 2
Overview

Ecommerce Analytics Dashboard is a cornerstone of modern business strategy. That focuses on developing and evaluating an e-commerce dashboard mobile application designed to help business owners monitor sales performance and analyze customer behaviour when it comes to sales effectively. The application integrates modern technologies such as Data Analytics Tools, Java, and Android Studio(IDE) to automate the tracking, overseeing, and analyzing customer information when it comes to sales and business processes.

The E-Dash application provides an interactive and user-friendly platform that enables business owners to view real-time insights into product movement, identify top-selling items, and monitor sales trends on a daily, monthly, and yearly basis. Key features include a Top Product Dashboard, Product History display, and Sales and Inventory Monitoring, which collectively enhance operational efficiency and build business trust.

Using both qualitative and quantitative research methods such as interviews, observation, and user testing the study evaluated the system’s usability, functionality, and overall impact on business operations. Results revealed that the E-Dash system significantly improves decision-making, strengthens customer relationships, and increases sales trends by providing accurate insights.

In conclusion, E-Dash demonstrates the potential of integrating data analytics and mobile technology to empower small and medium businesses. It offers an efficient solution for understanding customer behaviour, managing sales, and optimizing inventory processes, enhancing business performance in the e-commerce industry.

Traditional vs. Modern E-commerce Dashboard

Traditional E-commerce Dashboards often rely on basic metrics such as total sales, and order summaries. While useful, these dashboards often have particular qualities needed to provide actionable insights. Manual data compilation and limited integration capabilities further hinder their effectiveness.

In contrast, Modern dashboards implement technology such as providing insights in customers preferences, purchase history to tailor marketing, top products offerings , and overall customer experience, analyze and track inventory levels , stock performance to efficient operations and reduce cost.

Case Study
Review of Related Literature

This chapter presents existing studies and literature related to Ecommerce Dashboards. It discusses inventory management technologies, ecommerce dashboards, and the integration of databases and automation. The review supports the importance of implementing an Ecommerce Analytics Dashboards for efficient and accurate tracking and analyzing of their business processes and sales behaviours when it comes to sales.

E-Commerce Sales Analysis Dashboard using Business Intelligence Tool - Microsoft Power BI

According to Microsoft Power BI - R.Gokulpriya IJIRMPS Volume 12 (2024) the paper investigates on how Microsoft Power BI has affected a business intelligence solution that provides an analysis of e-commerce data extracted from sales for more robust business decision-making. The primary contributor to this cloud-based data warehouse is on AWS Redshift. Next step after gathering data into the Data-Warehouse is to extract some analytical data from that. This will increase income and lead to efficiency by improving business skills. In this study we are looking at three important aspects of business intelligence, business intelligence, data The study focuses on the use of business intelligence methods and technology in e-commerce sales analysis. It compares Power BI and Tableau, two popular data visualization technologies, and their effectiveness in predicting sales performance. The research highlights the

importance of data visualization and business intelligence in optimizing e-commerce sales analysis. The dashboard provides insights into sales, finance, marketing, and supply chain, enabling businesses to make informed decisions, improve resource allocation, and explore growth opportunities. It also uses advanced algorithms to provide accurate estimates of revenue, expense and profitability."

Microsoft Power BI - R.Gokulpriya - IJIRMPS Volume 12, Issue 4, July-August 2024. DOI 10.37082/IJIRMPS.v12.i4.230837

Real-time data analytics in retail: A review of USA and global practices

"Real-time data analytics has emerged as a pivotal tool in the retail sector, revolutionizing decision-making processes and operational strategies. This study delves into the extensive landscape of real-time data analytics in retail, offering a comprehensive review of practices both in the USA and on a global scale. In the United States, retail giants have harnessed real-time data analytics to gain unprecedented insights into consumer behaviour, preferences, and market trends. From personalized marketing campaigns to dynamic pricing strategies, retailers leverage real-time analytics to optimize inventory management and enhance the overall customer experience. The integration of technologies like RFID, IoT devices, and advanced analytics platforms has facilitated the seamless flow of data, enabling retailers to respond swiftly to changing market

dynamics. On a global scale, diverse retail markets have embraced real-time data analytics to stay competitive in an ever-evolving landscape. Cross-border e-commerce and the proliferation of online marketplaces have intensified the need for real-time insights into global consumer trends. Retailers worldwide are adopting predictive analytics to forecast demand, reduce under-stock products, and streamline supply chain operations. Moreover, the integration of artificial intelligence and machine learning algorithms enhances the ability to analyze vast datasets, uncovering hidden patterns and predicting future market trends. Challenges such as data privacy concerns, integration complexities, and the need for skilled professionals are acknowledged in this review. Despite these challenges, the transformative impact of real-time data analytics on the retail sector is undeniable. As technology continues to advance, the synergy between real-time data analytics and retail practices is expected to deepen, fostering innovation and reshaping the industry's landscape. This study provides a glimpse into the dynamic and evolving realm of real-time data analytics in retail, offering insights into the strategies employed by both the USA and global players to thrive in an era of rapid technological change.”

Raji, M. A., Olodo, H. B., Oke, T. T., Addy, W. A., Ofodile, O. C., & Oyewole, A. T. (2024). Real-time data analytics in retail: A review of USA and global practices. *GSC Advanced Research and Reviews*, *18*(3), 059-065.

LiveRetro: Visual Analytics for Strategic Retrospect in Livestream E-Commerce

"Livestream e-commerce integrates live streaming and online shopping, allowing viewers to make purchases while watching. However, effective marketing strategies remain a challenge due to limited empirical research and subjective biases from the absence of quantitative data. Current tools fail to capture the interdependence between live performances and feedback. This study identified computational features, formulated design requirements, and developed *LiveRetro*, an interactive visual analytics system. It enables comprehensive retrospective analysis of livestream e-commerce for streamers, viewers, and merchandise. *LiveRetro* employs enhanced visualization and time-series forecasting models to align performance features and feedback, identifying influences at channel, merchandise, feature, and segment levels. Through case studies and expert interviews, the system provides deep insights into the relationship between live performance and streaming statistics, enabling efficient strategic analysis from multiple perspectives."

Wu, Y., Xu, Y., Gao, S., Wang, X., Song, W., Nie, Z., ... & Li, Q. (2023). LiveRetro: Visual analytics for strategic retrospect in livestream e-commerce. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 1117-1127.

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Microsoft Power BI - R.Gokulpriya - IJIRMPS Volume 12, Issue 4, July-August 2024. DOI 10.37082/IJIRMPS.v12.i4.230837.

Raji, M. A., Olodo, H. B., Oke, T. T., Addy, W. A., Ofodile, O. C., & Oyewole, A. T. (2024). Real-time data analytics in retail: A review of USA and global practices. *GSC Advanced Research and Reviews*, *18*(3), 059-065.

Wu, Y., Xu, Y., Gao, S., Wang, X., Song, W., Nie, Z., ... & Li, Q. (2023). LiveRetro: Visual analytics for strategic retrospect in livestream e-commerce. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 1117-1127.

Chapter 3

Methodology

This chapter outlines the research methodology that will be deployed in the development of **"E-commerce Analytics Dashboard"**. The primary goal of this research is to design and implement an E-commerce Dashboard that automates the tracking, monitoring of customer data and analyzing the sales and needs of the customers for business processes. This chapter will detail the research approach, data collection methods, system design, tools, and techniques that will be used to achieve this goal.

The research will adopt a systematic approach to provide the effective development and implementation of the proposed application. Initially, a complete review of the existing E-commerce Dashboard will be conducted to identify the gaps and areas for improvement. Based on this review, the methodology will proceed with the design phase, where the components of the E-commerce Analytics Dashboard will be defined. To gather some valuable insights on user requirements and expectations, surveys and interviews will be conducted with key respondents, including industry professionals and experts in the field of E-commerce Dashboards. These respondents will play a crucial role in providing the necessary data for the application's design and functionality. Their information and feedback will be the guide in the development process, ensuring the applications meet the practical needs of users.

Research and Development Stage

The Research and Development stage is a critical phase in the creation of the "E-commerce Analytics Dashboard." During this stage, the development process will be guided by the insights and feedback collected from the respondents, including industry professionals and experts. These respondents will provide valuable input on the system's design, functionality, and customers needs, ensuring that the final product is practical and meets the requirements of E-commerce Dashboard.

The responses will help to identify and determine the key challenges faced by the e-commerce owners and highlight some areas where the new application can bring some improvements, such as automation, clear customer insights tracking, and more accurate data reporting. Based on this research, the respondents will proceed with the design of the application. The system architecture will include some modern technologies such as Google Analytics Tools for clearer insights into user's behaviours and sales data, Data Visualization Tools for creating interactive and visually appealing dashboards and Microsoft Excel for leveraging their data visualization features. The respondents' feedback will influence the features and interfaces of the application to align with the user's expectations and business operations needs. During the development phase, the application will be built in stages, with regular testing and evaluation to meet performance standards. Throughout this process, additional feedback from respondents will be incorporated to refine and improve the application. Once the

prototype is completed, further user testing will be conducted to make the application intuitive,user- friendly, and efficient in managing customer data and analyzing sales and needs tasks.

We Use Agile Development Methodology

The Agile Methodology is a structure we used in the development of the "E-commerce Analytics Dashboard Application". The application will follow the Agile methodology as the type of methodology used to develop a project that is completed systematically, efficiently, and effectively. The Agile Methodology will guide the development process from the initial planning phase to the final deployment, ensuring that the application meets the needs of the customers and users that aligns with the insights provided by the respondents.

The Agile will consist of the following stages:

1.Planning

The first phase of the Agile methodology involves gathering detailed requirements for the system. In this stage, the input from the respondents, including e-commerce owners will be essential in identifying the specific challenges and needs that the application must address. Through surveys, interviews, and discussions, the researchers will gather some specific data on the key features, desired functionalities, and the technical requirements of the ecommerce dashboard application. This phase will also involve setting clear goals, defining the project scope, and developing a timeline for the project.

2. Application Design

In the design phase, the applications architecture and overall structure will be defined. The respondents' feedback will guide the design of both the front-end (user interface) and back-end (server and database structure). The design will also include modern technologies such as Google Analytics Tools, Data Visualization Tools, data analytics, and Microsoft Excel and Google Sheets ensuring that the application is both scalable and user-friendly. The design phase will focus on ensuring that PHINMA UNIVERSITY OF ILOILO College of Information Technology Education the application meets the functional and non-functional requirements identified in the previous phase.

3. Development

During the development phase, the actual coding and implementation of the application will take place. Based on the designs, the developers will create the software using

appropriate programming languages and tools. The respondents will continue to play an important role by reviewing early prototypes and providing some feedback on features such as usability, accuracy, and performance. This repeating process will help the application be aligned with user and customer needs and expectations.

4. Testing

Once the application is developed, Testing will involve both functional testing (to check if all features are working correctly) and performance testing (to test if the application can handle large amounts of data and usage). The respondents will be involved in user acceptance testing (UAT), where they will test the application to identify any issues or areas for improvement. Feedback from this phase will help improve the application before its full deployment.

5. Deployment

After successful testing, the application will be deployed in a live environment. This phase involves configuring the application for the actual use, ensuring that it is fully operational and accessible to users. The deployment phase will also involve providing initial training to users, so they can effectively operate the application. Respondents will be consulted during the deployment process.

6. Review

The final phase is conducting a final review of the application's performance and user satisfaction. Feedback from users and respondents will be collected to evaluate whether the application meets the specific goals, expectations and identify potential updates or enhancements for the application.

Data Collection Techniques

In order to gather the necessary insights for developing the "E-commerce Analytics Dashboard Application," a variety of data collection techniques will be implemented. These techniques will help the system meet the needs of customers, particularly E-commerce owners and other stakeholders, while also addressing the current challenges in E-commerce Dashboard.

The following data collection methods will be used

1.Surveys

Surveys will be used as a primary method for collecting and gathering quantitative data from a wide range of respondents, including e-commerce owners, business managers, etc. The surveys will be designed to capture information on their current e-commerce dashboard practices, challenges, and expectations for an improved application. The survey questions will be structured to gather both demographic data and specific feedback on the key features, application requirements,and performance

expectations. This will provide valuable insights into the features that should be prioritized in the e-commerce analytics dashboard.

2.Observations

Observations will be conducted to understand the daily operations of E-commerce Dashboard in real-world settings. This technique involves observing dashboards processes in action at different organizations to identify inefficiencies, and some areas that could benefit from automation .

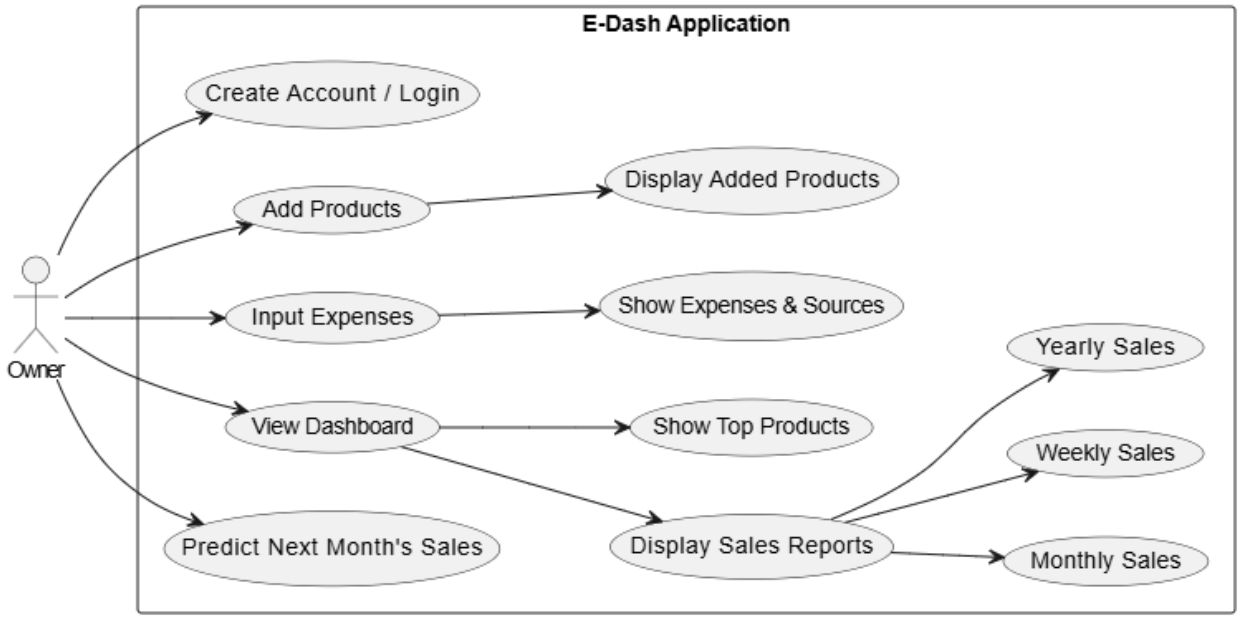
Justification for Selecting the Agile Methodology

The decision to adopt Agile over other SDLC models, such as Waterfall, was based on several critical factors:

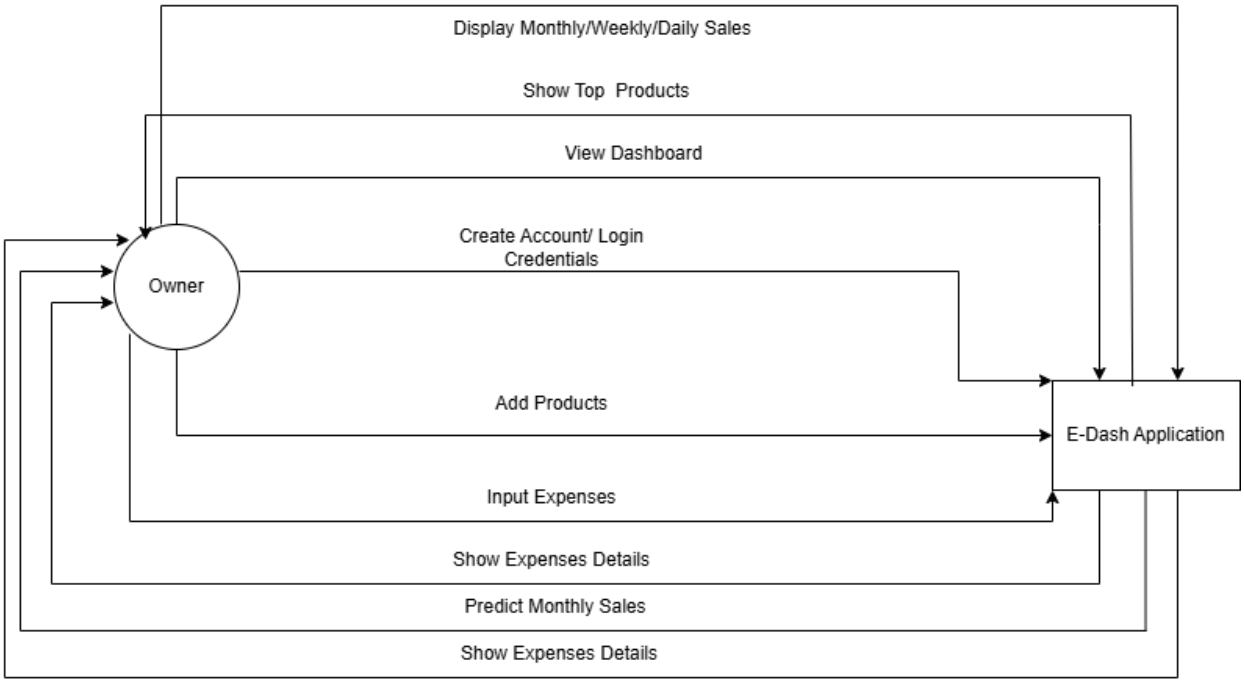
Criteria	Agile Methodology	Waterfall Model
Requirement Gathering	Continuous input from respondents (e-commerce owners, users) through observations, interviews, and feedback	Requirements fixed at the start; little to no user involvement after documentation

Design Approach	Iterative design guided by respondent feedback, focusing on scalability and usability	Rigid design completed before development; hard to adjust to changing requirements
Development Process	Incremental coding with early prototypes reviewed by respondents	Full system developed before user evaluation
Testing	Ongoing testing in each iteration, plus User Acceptance Testing (UAT) with respondents	Testing occurs only after full development, increasing risk of late issue discovery
Deployment	Gradual release with training and respondent consultation to ensure usability	Single release after project completion, minimal user preparation or feedback
Review & Improvement	Post-deployment reviews and feedback loops for future updates and enhancements	Rarely includes structured feedback; updates require separate new projects

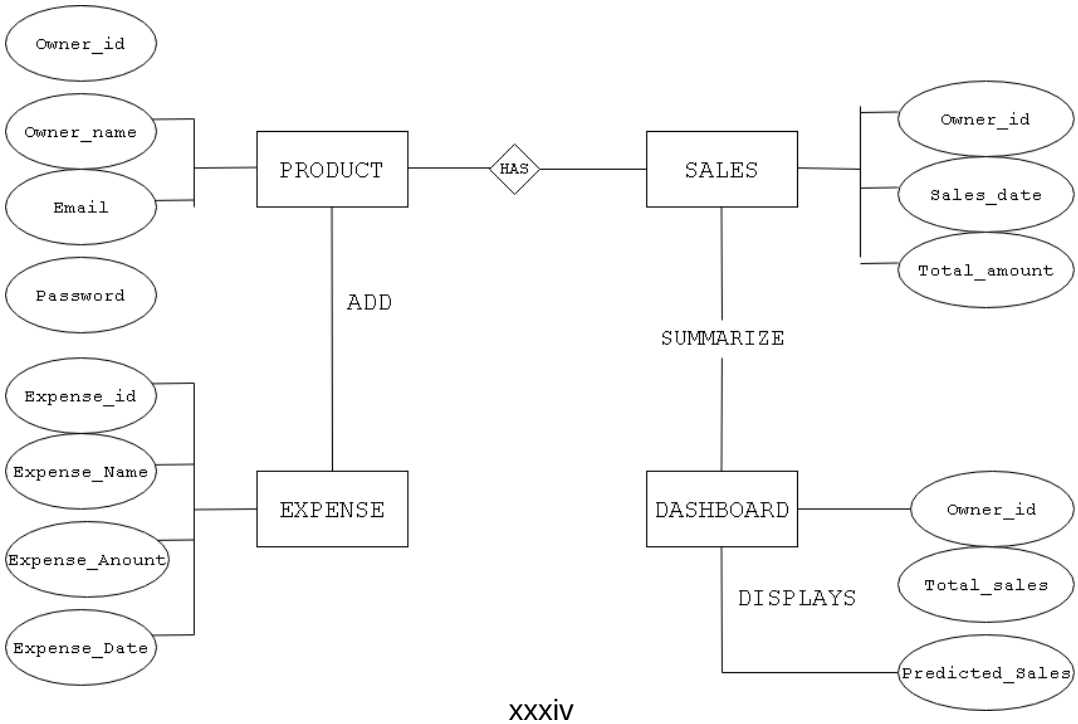
Use Case Diagram



Data Flow Diagram



Entity Relationship Diagram



Survey Questionnaire for System Assessment

This survey aims to gather information from business owners to better understand the challenges that they faced and guides us in creating a platform that helps us in developing an E-commerce Analytics Dashboard that provides real-time tracking, data visualization, and to enhance efficiency and effective decision-making for small to medium enterprises (SMEs) in the e-commerce industry.

What is your Name ?

What is your age range?

- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old

What type of business do you own?

- Retail
- Services
- Wholesale
- Manufacturing
- Other

How long have you been operating your business?

- Less than a year

- 1-3 years
- 4-6 years
- 7 years above

Do you currently use tools for sales tracking?

- Yes
- No

Do you use spreadsheets or reports for sales tracking?

- Yes
- No

Do you update your sales tracking data daily?

- Yes
- No

Do you update your sales tracking data daily?

- Yes
- No

Do you use data analytics tools to monitor customer behavior?

- Yes
- No

Do you use cloud storage for your business data?

- Yes
- No

How challenging is it for you to analyze customer behavior data?

- Very Easy
- Moderate
- Hard

Would you use a mobile application to monitor your business operations?

- Yes
- No

Do you need access to real-time analytics daily?

- Yes
- No

Do you find your existing analytics tools effective in understanding customer behavior?

- Yes
- No

Do you find it effective to use a mobile application for analytics?

- Yes
- No

Do you feel your current analytics tools meet your business needs?

- Yes
- No

Do you believe analytics dashboards are effective in the cost?

- Yes
- No

Would you be motivated to use an E-commerce Analytics Dashboard?

- Yes
- No

Would you prefer automated reports over manual reports?

- Yes
- No

Do you store your business data securely ?

- Yes
- No

Is analyzing customer data a challenge for you?

- Yes
- No

Chapter 4

Results and Interpretation

This chapter presents the complete outcomes of the development, implementation, and evaluation of Ecommerce Analytics Dashboard: The modern ecommerce dashboard - based on its ability to track sales trends, display top-performing products, and provide real time data insights. After implementation, results were gathered through system performance tests, user testing. The results are systematically organized to demonstrate how the developed system addresses the identified problems and achieves the established objectives. The findings are based on extensive data collection from the owner of Sme’s, potential customers, system performance metrics, and user acceptance testing conducted throughout the planning and deployment phases.

Systems and Features

E-dash is a mobile application that helps business owners to track their sales and analyze their customer’s sales behavior this systems features include:

1. **Top Products Dashboard**
   - Displays the best performing products based on sales within their selected time frames daily, weekly, monthly, or yearly.

2. **Monthly Sales Forecasting**
   - Predicts future revenue by analyzing past sales data, market trends
3. **Sales Inventory Management**
   - Monitors stock levels to prevent overstocking or under stocking.
4. **Daily Product Sales Report**
   - Provides a summary of daily sales report
5. **Owner's Income Dashboard**
   - Provides an overview of the owner's total income within a selected Day, Week, Month, and a Year.

Result

All system features were successfully developed and tested. The Owners were able to monitor their sales easily, complete sales overview, and accurate inventory management. The sales marketing strategies effectively increased engagement and sales intent during the testing and deployment phase.

Implementation Challenges

The data used for analysis in this study was gathered through various sources, including system logs, inventory transaction records, and user testing sessions. Primary data sources include the E-commerce Analytics Dashboard backend database, which recorded all item movements, stock-in and stock out, and user interactions during the application testing period. Additional data was obtained through surveys and interviews with the user(owner) to gather feedback and insights on system usability and efficiency.

For analysis, both quantitative and qualitative methods were applied. Quantitative data, such as transaction accuracy and inventory differences before and after system implementation, were analyzed to measure performance improvements. Qualitative feedback from users was categorized and interpreted to assess system functionality, and areas for improvement. Together, these methods provided a complete understanding of how well the application performs in a real-world scenarios.

User Feedback

Based on the observation on deploying this application the user has been satisfied with the features of this application. Business owners expressed high satisfaction with the E-Dash application, appreciating its ability to provide clear insights into their sales performance and customer behavior. They were impressed with the interactive and organized design of the dashboard, which made it easy for them to monitor product movement, income, and top-selling products. Owners stated that the system helped them track their daily, monthly, and yearly sales efficiently, giving them a better understanding of their business trends. The owner shared that the Sales Analytics and Owner’s Income Dashboard features motivated them to make informed business decisions and manage their inventory more effectively. Additionally, respondents found the user interface was simple, visually appealing, and easy to navigate. However, some recommended further improving the system’s loading speed and performance optimization to ensure smoother usage, especially when handling large sales data.

Comparison to other System

Compared to other mobile applications, E-Dash aims to meet the needs of the owner by providing easy to use features. While the E-Commerce Sales Analysis Dashboard using Business Intelligence Tool Microsoft Power BI focuses on analyzing large-scale e-commerce data through cloud-based infrastructures and advanced visualization tools (Gokulpriya, 2024), E-Dash offers a more accessible and user-centered approach. Instead of relying on cloud storage, E-Dash uses a local SQLite database that allows users to manage and analyze their data directly on mobile devices.

E-Dash also integrates features such as expense tracking, stock in and stock out monitoring, sales history viewing, and sales prediction for the upcoming month. These additional features make E-Dash reliable and user-friendly, especially for small business owners who need a simple yet powerful tool to analyze sales performance and make informed business decisions anytime and anywhere.

Chapter 5

Summary, Conclusion, and Recommendation

Summary

This study aims to design and develop E-Dash "Analyzing Customer’s Sales Behaviour and Tracking Business Process", a system that analyzes customers behaviour when it comes to sales and tracking business processes , provides real-time inventory information, and offers top selling products based on their total sales. The platform includes features such as weekly and monthly sales inventory, top products dashboard, sales inventory management, and owner’s income dashboard features to enable business owners to improve efficiency, increase profitability, and make more informed business decisions. . The project followed a structured development process, including planning, application design, development, testing, and review.

The results showed significant improvements in E-Dash efficiency and user satisfaction. The system successfully met its objectives and proved to be an effective solution for helping business owners to analyze their customer’s sales behaviour and track business process easily and efficiently.

Conclusion

The project E-Dash: Analyzing Customer’s Behaviour and Tracking the Business Process was developed to address business owner’s struggles, without an efficient approach, decision-

making processes become reactive rather than proactive, leading to inefficiencies in inventory management, sales monitoring, understanding customers' insights and customer relationships. These issues make it hard for business owners to earn more sales income, monitor fast movement products based on their selected sales , and plan their purchase activities.

The E-Dash system was able to solve that problem by providing accurate insights based on sales. Owners will directly know their monthly sales income based on the application features. Owners can easily view available products, monitor sales information. It also provides real time and accurate data analytics and top sales product recommendation based on the total sales, helping owners make better decisions-making.

By connecting customers directly to owners, the system helps reduce the owners' reliance on manual tracking their products movements and sales.

Recommendations

Based on the development and evaluation of the project, the following recommendations are made:

1. Sales Analytics

This feature provides business owners with real-time insights into sales performance. It displays visual analytics such as graphs and charts that summarize total sales, sales growth, and customer buying patterns. By analyzing sales trends, users can make informed decisions to improve marketing strategies and business operations.

2. Inventory Management

The Inventory Management feature allows users to track product stocks, monitor product movement, and update stock in and stock out records. It helps prevent overstocking or stock shortages by providing accurate, up-to-date inventory information, ensuring efficient inventory control.

3. Monthly Sales

This feature provides a detailed summary of monthly sales performance. Business owners can view total revenue, compare current month sales with previous periods, and identify growth patterns.

4. Top Products Dashboard

The Top Products Dashboard displays the best-selling products based on sales data within a selected time frame (daily, weekly, monthly, or yearly).

5. Owner's Income Dashboard

This feature presents the total income earned by the business owner over specific periods. It visualizes income data through graphs and summaries, allowing users to assess profitability, track earnings, and plan for business growth effectively.

6. Daily Product Sales Report

The Daily Product Sales Report generates a summary of all products sold each day. It allows business owners to review daily sales transactions, identify high-demand products, and monitor daily revenue.

Final Thoughts

The development of E-Dash Analyzing Customers Sales Behaviour and Tracking Business Process demonstrates how technology can simplify business management and support data driven decision-making. By providing real-time sales analytics, product tracking, and sales monitoring, the system helps business owners gain better insight into their sales performance and overall operations. Through this project, the team realized that innovative yet simple technological solutions can enhance business efficiency and profitability. E-Dash is not just a business tool, it is a step toward empowering business owners, improving productivity, and promoting more efficient business practices.

Reference (APA Format)

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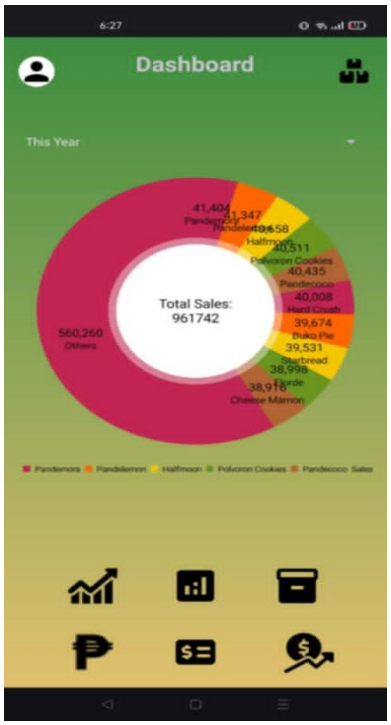
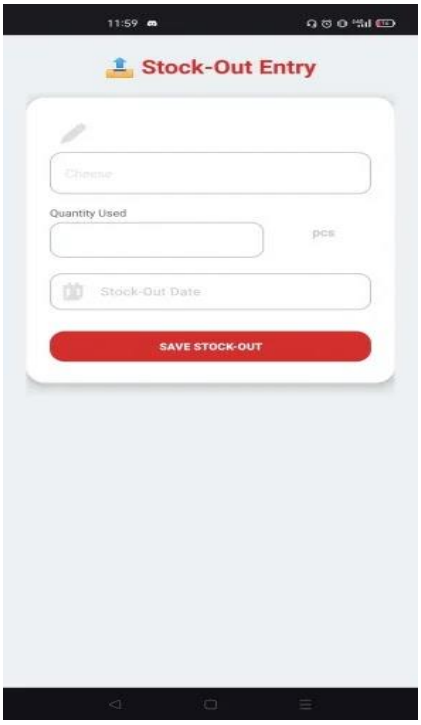
2.Wu, Y., Xu, Y., Gao, S., Wang, X., Song, W., Nie, Z., ... & Li, Q. (2023). LiveRetro: Visual analytics for strategic retrospect in livestream e-commerce. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 1117-1127.

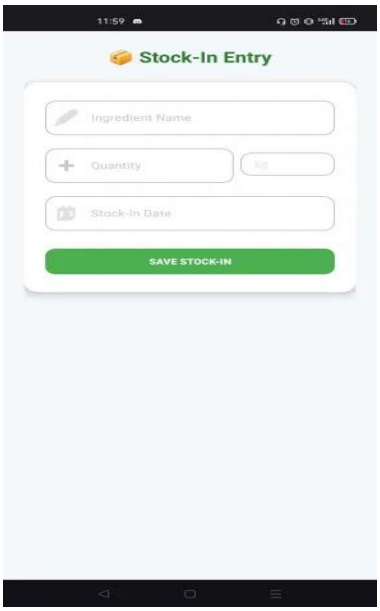
3.Microsoft Power BI - R.Gokulpriya - IJIRMPS Volume 12, Issue 4, July-August 2024. DOI 10.37082/IJIRMPS.v12.i4.230837

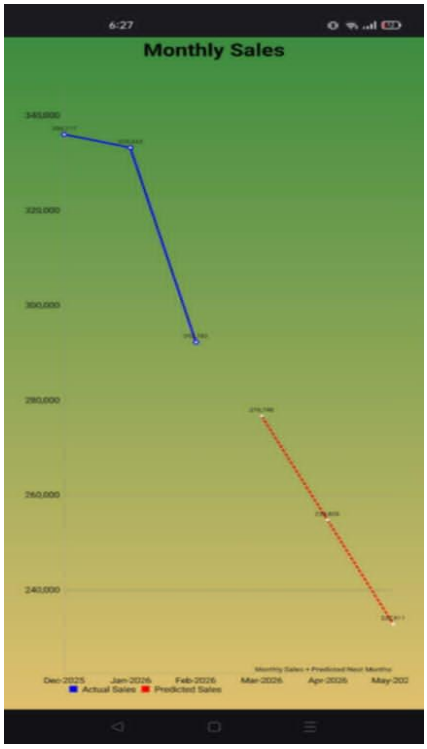
Appendices

Screenshots





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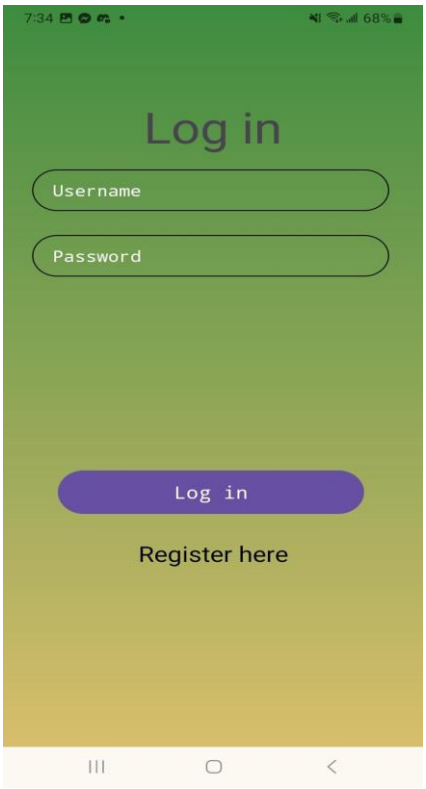
Monthly Sales
Actual Sales
Predicted Sales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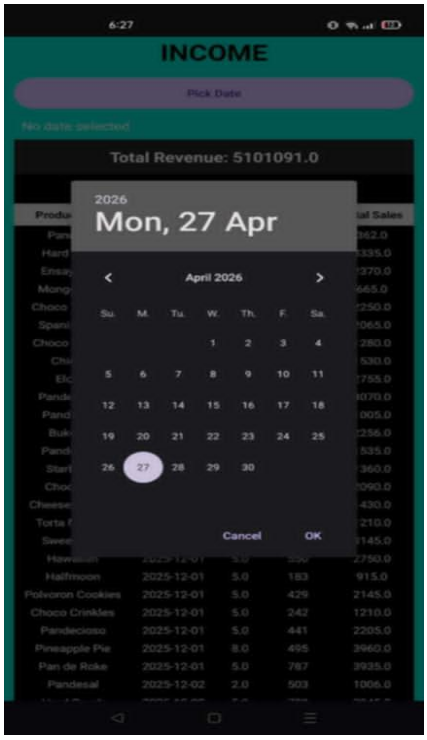
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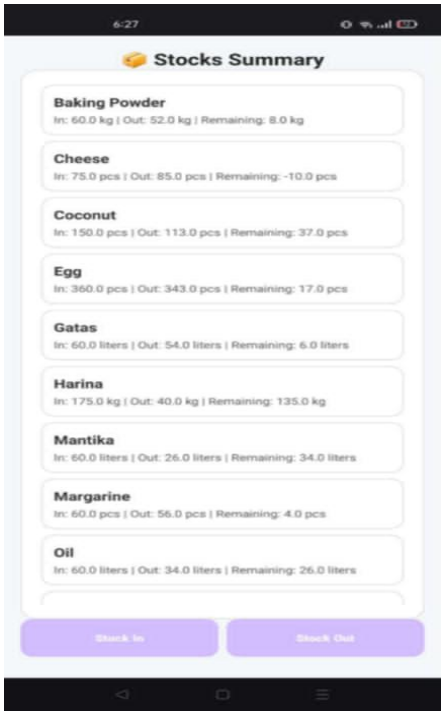
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Stocks Summary
Baking Powder
In: 60.0 kg | Out: 52.0 kg | Remaining: 8.0 kg
Cheese
In: 75.0 pcs | Out: 85.0 pcs | Remaining: -10.0 pcs
Coconut
In: 150.0 pcs | Out: 113.0 pcs | Remaining: 37.0 pcs
Egg
In: 360.0 pcs | Out: 343.0 pcs | Remaining: 17.0 pcs
Gatas
In: 60.0 liters | Out: 54.0 liters | Remaining: 6.0 liters
Harina
In: 175.0 kg | Out: 40.0 kg | Remaining: 135.0 kg
Mantika
In: 60.0 liters | Out: 26.0 liters | Remaining: 34.0 liters
Margarine
In: 60.0 pcs | Out: 56.0 pcs | Remaining: 4.0 pcs
Oil
In: 60.0 liters | Out: 34.0 liters | Remaining: 26.0 liters
Stock In
Stock Out

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